Workflow and architecture patterns for working together 2: Architectures

Jacob Archambault

June 2, 2026

Outline

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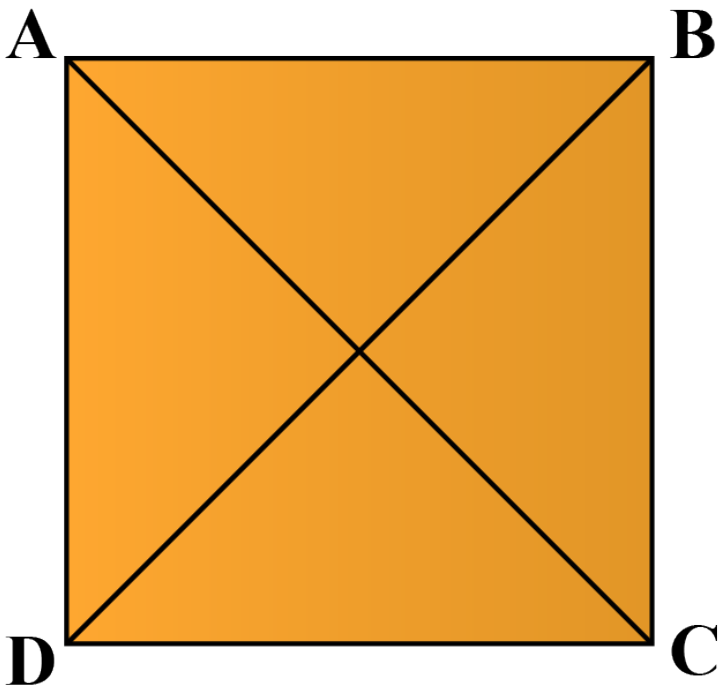
1975: The Mythical Man Month, Fred Brooks

- number of direct communication paths for n individuals= $n(n-1)/2$



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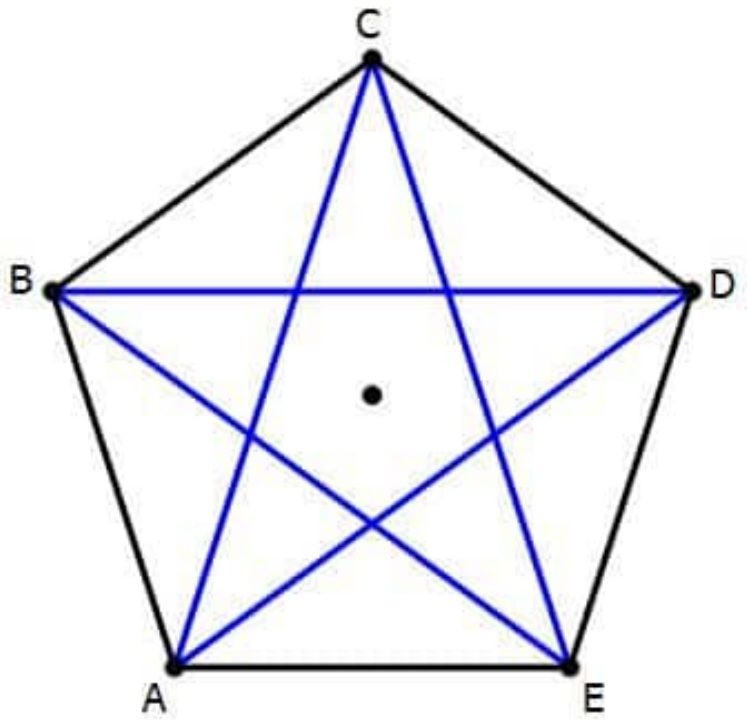

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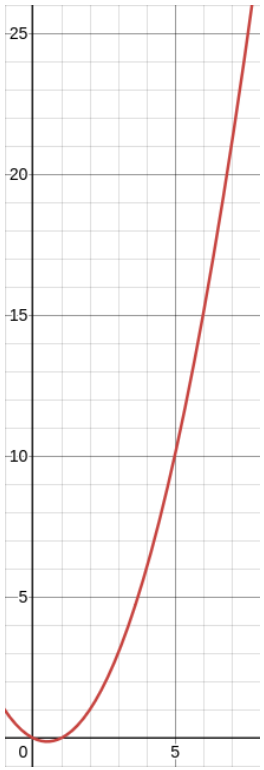
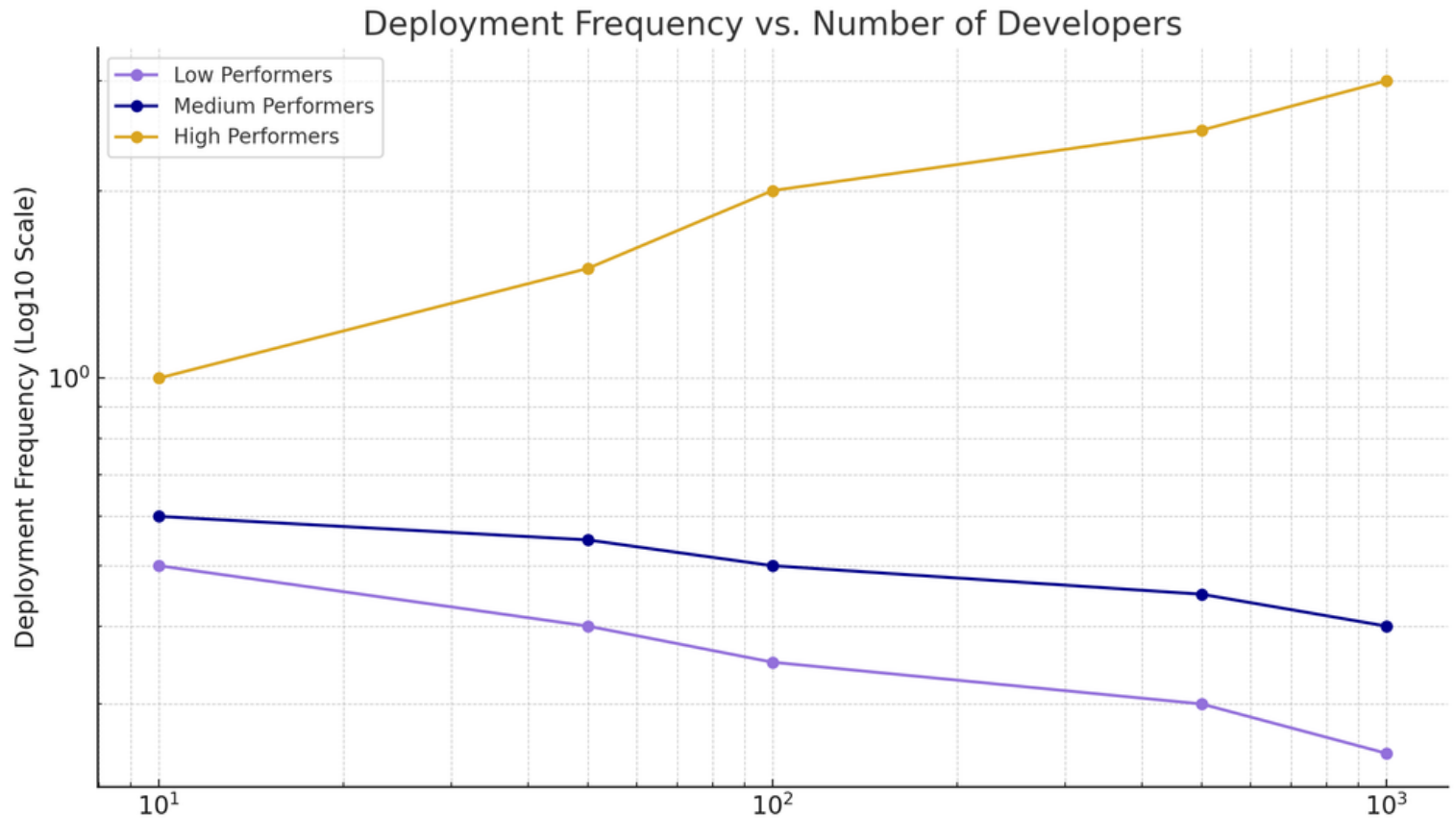


The Mythical Man Month, Fred Brooks (cont.)

- corollary: adding more people to a project can lead not only to diminishing returns on delivery speed, but to objectively less work being completed



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1967: Conway's law

'Organizations which design systems [...] are constrained to produce designs which are copies of the communication structures of these organizations.' - Melvin Conway, 'How do Committees Invent?' Datamation, 1967

Conway's law: examples

- separate `.csproj` projects for tests written by QA
- separate development and config repositories for the same transaction signifying low communication between development and operations
- all business documentation in the same repository, separate from the repositories that the documentation is for

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  - branching-based strategies for team management compromise between the above approaches without their benefits

Revisiting Conway's law: defining a team

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A naive probability theory for merge consistency: isolated commits

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A naive probability theory for merge consistency: general definition for distinct branches diverging from a common ancestor

- Let A be the set of unmerged commits across k distinct branches $A_1 \cdots A_k$, where $A = A_1 \sqcup A_2 \sqcup \cdots \sqcup A_k$, $n = |A|$, and $n_i = |A_i|$ for $1 \leq i \leq k$. Then

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$$P(\textit{no conflicts}) = \frac{1}{2^n - (\sum_{i=1}^{k} 2^{n_i}) + k}$$

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The State of DevOps Report: throughput metrics

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The State of DevOps Report: stability metrics

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The State of DevOps Report: results

'Astonishingly, these results demonstrate that there is no trade-off between improving performance and achieving higher levels of stability and quality. Rather, high performers do better at all of these measures. This is precisely what the Agile and Lean movements predict, but much dogma in our industry still rests on the false assumption that moving faster means trading off against other performance goals, rather than enabling and reinforcing them.'
- Nicole Forsgren, Jez Humble, and Gene Kim, Accelerate (2017), p. 14

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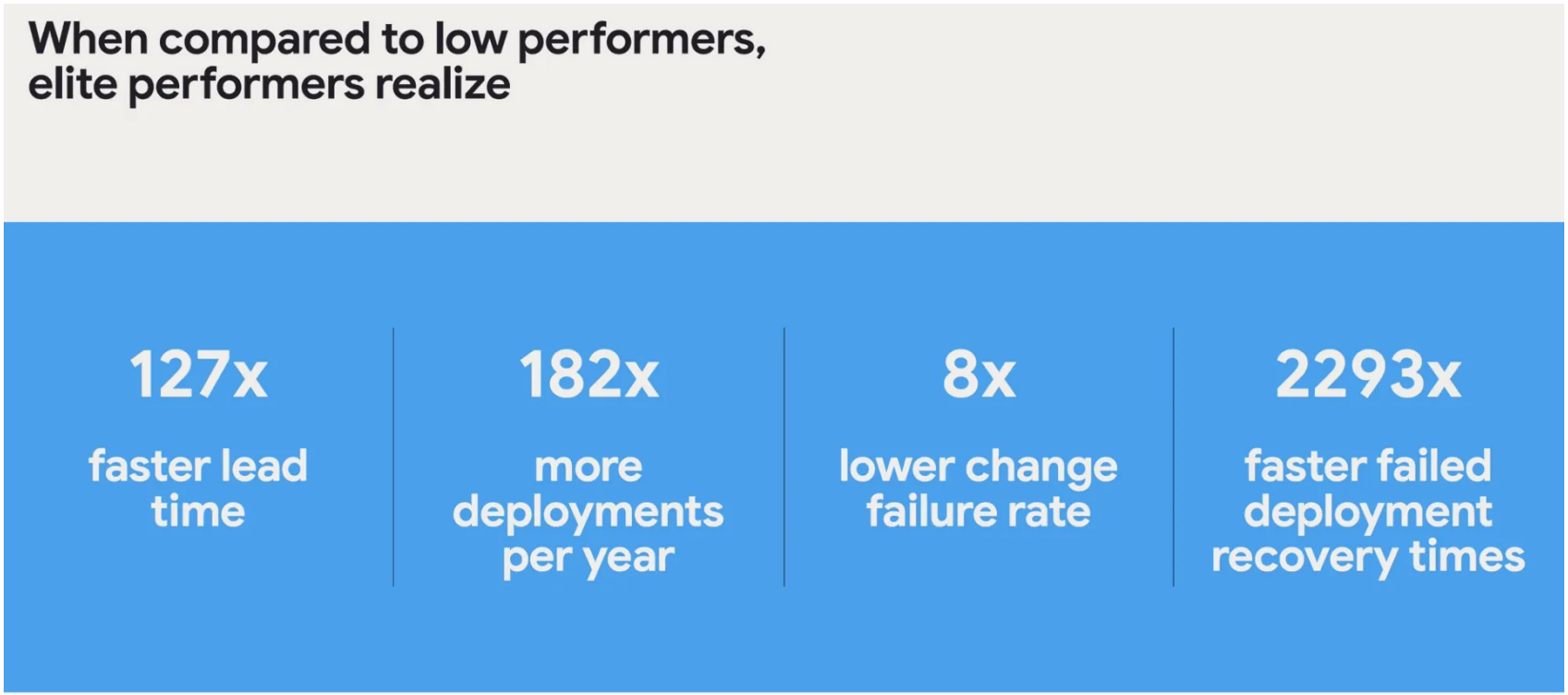

- 2024 State of DevOps Report

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Branch	Runs	Earliest run	Contributors
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Either COB branch	159	23 Mar	6
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Source: GH Actions Simulation tests, 28 May 2026

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Either COB branch	76/159 (48%)	77/159 (48%)	23 Mar
All extant branches	73/123 (59%)	87/123 (71%)	23 Mar
	50/72 (69%)	51/72 (71%)	15 Apr
	92/161 (57%)	104/161 (64%)	21 Jan

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Source: GH Actions, 28 May 2026

- 28% better simulation test pass rate since 23 Mar
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Stability: pipeline failure rate metrics

Branch	Simulation test failures	Unit test failures	Earliest commit
NG_Claims_837D_COB	56/74 (76%)	56/74 (76%)	23 Mar
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Either COB branch	76/159 (48%)	77/159 (48%)	23 Mar
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- Source command: `git log --since="2026-04-15" --oneline <--merges> <--all> <^PEDI-35120_837D_COB> | <grep -iv ''merge pull request''> | wc -l`

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